



Evasion Defenses

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Today's Agenda

1 Adversarial Training

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Towards Deep Learning Models Resistant to Adversarial Attacks

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Optimization View on Adversarial Robustness

- The goal of standard training is to find model parameters θ that minimize the risk function $\mathcal{L}$:

$$\min_{\theta} \mathbb{E}_{(x,y) \sim \mathcal{D}} [\mathcal{L}(f_{\theta}(x), y)]$$

where data distribution $\mathcal{D}$ is over pairs of examples $x \in \mathbb{R}^d$ and corresponding labels $y \in [K]$ (K is the number of classes).

- Unfortunately, **ERM often does not yield models that are robust** to adversarially crafted examples.

Robust Optimization View on Adversarial Robustness

In order to reliably train models that are robust to adversarial attacks, it is necessary to **augment the ERM paradigm** appropriately by changing the definition of population risk $\mathbb{E}_{\mathcal{D}}[\mathcal{L}]$

- Instead of feeding samples from the distribution $\mathcal{D}$ directly into the loss $\mathcal{L}$, we allow the adversary to **perturb the input first**.
- This gives rise to the following saddle point problem:

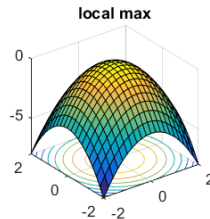
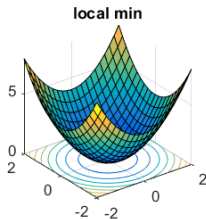
$$\min_{\theta} \mathbb{E}_{(x,y) \sim \mathcal{D}} \left[\max_{\delta \in \Delta} \mathcal{L}(f_{\theta}(x + \delta), y) \right]$$

where $\Delta \subseteq \mathbb{R}^d$ is allowed perturbations that formalizes the manipulative power of the adversary (e.g., ℓ_{∞} -ball).

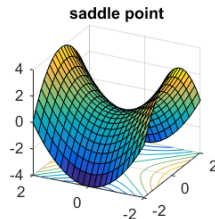
Adversarial Loss

Our perspective stems from viewing the **saddle point** problem as the composition of an **inner maximization** problem and an **outer minimization** problem.

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[Source](#)

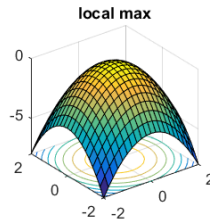
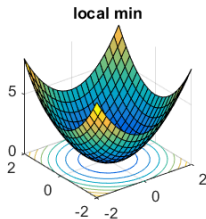


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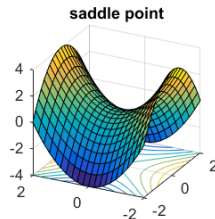
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- The **inner maximization** problem aims to **find an adversarial version of a given data point x** that achieves a high loss.



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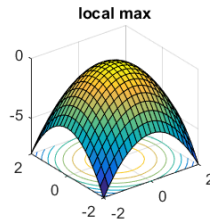
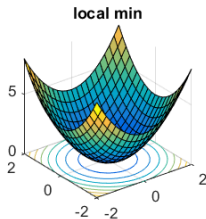


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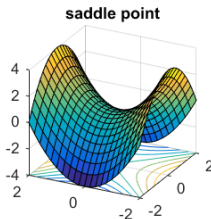
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- The goal of the **outer minimization** problem is to **find model parameters** so that the **adversarial loss** given by the inner attack problem is minimized.



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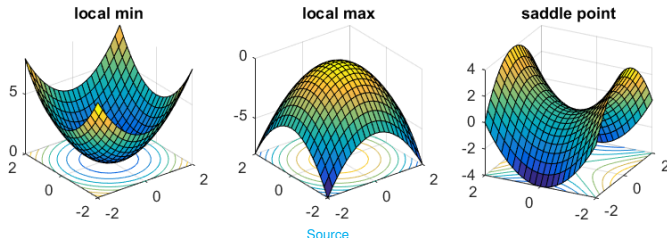


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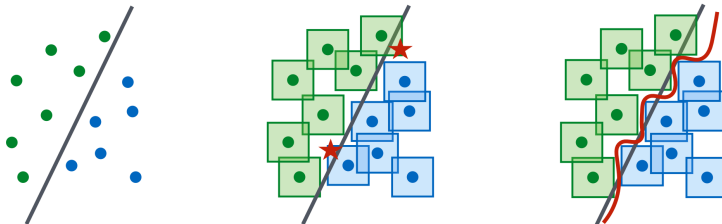
- The **inner maximization** problem aims to **find an adversarial version of a given data point** x that achieves a high loss.
- The goal of the **outer minimization** problem is to **find model parameters** so that the **adversarial loss** given by the inner attack problem is minimized.
- When the parameters θ yield a (nearly) **vanishing risk**, the corresponding model is perfectly **robust to attacks** specified by our attack model.



Network Capacity and Adversarial Robustness

Classifying examples in a robust way requires a stronger classifier, since the presence of adversarial examples changes the decision boundary of the problem to a more complicated one.

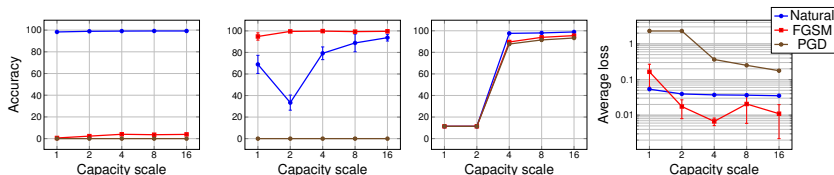
- Experiments verify that capacity is crucial for robustness, as well as for the ability to successfully train against strong adversaries.



Network Capacity and Adversarial Robustness

Either **increasing the capacity** of the network, or using a **stronger method for the inner optimization** problem reduces the effectiveness of adversarial inputs (in other words, **increase the robustness of model**).

MNIST



CIFAR10

	Simple	Wide	Simple	Wide	Simple	Wide	Simple	Wide
Natural	92.7%	95.2%	87.4%	90.3%	79.4%	87.3%	0.00357	0.00371
FGSM	27.5%	32.7%	90.9%	95.1%	51.7%	56.1%	0.0115	0.00557
PGD	0.8%	3.5%	0.0%	0.0%	43.7%	45.8%	1.11	0.0218
(a) Standard training			(b) FGSM training ¹			(c) PGD training		
(d) Training Loss								

¹ Why is FGSM accuracy is higher than Natural accuracy?

Adversarially Trained Models

MNIST

- 40 iterations of projected gradient descent
- $\alpha = 0.01$ and $\epsilon = 0.03$
- Train model on natural samples: 99.2% accuracy on validation set

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CIFAR10

- 7 iterations of projected gradient descent
- $\alpha = \frac{3}{255}$ and $\epsilon = \frac{8}{255}$
- Train model on natural samples: 95.2% accuracy on validation set

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Source models to generate adversarial examples

- A : White-box attack
- A' : Black-box attacks from an independently trained copy of the network
- A_{nat} : Black-box attacks from a version of the same network trained only on natural examples
- B : Black-box attacks from a different convolution architecture

Adversarially Trained MNIST Model

Method	Steps	Restarts	Source	Accuracy
Natural	-	-	-	98.8%
FGSM	-	-	A	95.6%
PGD	40	1	A	93.2%
PGD	100	1	A	91.8%
PGD	40	20	A	90.4%
PGD	100	20	A	89.3%
Targeted	40	1	A	92.7%
CW	40	1	A	94.0%
CW+	40	1	A	93.9%
FGSM	-	-	A'	96.8%
PGD	40	1	A'	96.0%
PGD	100	20	A'	95.7%
CW	40	1	A'	97.0%
CW+	40	1	A'	96.4%
FGSM	-	-	B	95.4%
PGD	40	1	B	96.4%
CW+	-	-	B _I	95.7%

Table 1: MNIST: Performance of the adversarially trained network against different adversaries for $\epsilon = 0.3$. For each model of attack we show the most successful attack with bold. The source networks used for the attack are: the network itself (A) (white-box attack), an independently initialized and trained copy of the network (A'), architecture B from [29] (B).

Adversarially Trained CIFAR10 Model

Method	Steps	Source	Accuracy
Natural	-	-	87.3%
FGSM	-	A	56.1%
PGD	7	A	50.0%
PGD	20	A	45.8%
CW	30	A	46.8%
FGSM	-	A'	67.0%
PGD	7	A'	64.2%
CW	30	A'	78.7%
FGSM	-	A_{nat}	85.6%
PGD	7	A_{nat}	86.0%

Table 2: CIFAR10: Performance of the adversarially trained network against different adversaries for $\epsilon = 8$. For each model of attack we show the most effective attack in bold. The source networks considered for the attack are: the network itself (A) (white-box attack), an independently initialized and trained copy of the network (A'), a copy of the network trained on natural examples (A_{nat}).

Adversarial Training

We find that **adversarial training alleviates the texture bias** of standard CNNs when trained on object recognition tasks, and helps CNNs **learn a more shape-biased representation**. This finding partially explains why adversarially trained-CNNs tends to be more robust than standard CNNs.

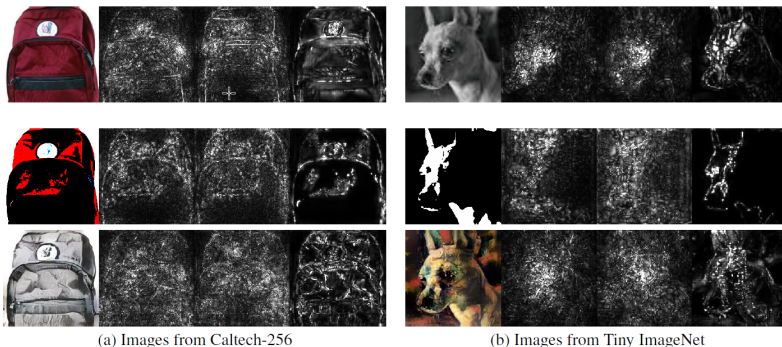


Figure 2. Sensitivity maps based on SmoothGrad (Smilkov et al., 2017) of three models on images under saturation, and stylizing. From top to bottom, Original, Saturation 1024 and Stylizing. For each group of images, from left to right, original image, sensitivity maps of standard CNN, underfitting CNN and PGD- l_∞ AT-CNN.

References and Resources I

- Adversarial Training (AT) was first introduced by Goodfellow et al. [5], however the adversary utilized was quite weak. Madry et al. [7] were the first to use the PGD attack to improve their results. As a consequence of their wildly successful paper, their method of Adversarial Training is now considered to be the standard version.
- Athalye, Carlini, and Wagner [1] identified obfuscated gradients and broke 7 out of 9 defenses [2, 6, 4, 9, 8] from ICLR 2018. Only Adversarial Training [7] managed to retain its performance and as a result, emerged as the most widely used defense mechanism against adversarial examples.
- Zhang et al. [10] identify a trade-off between robustness and accuracy that serves as a guiding principle in the design of defenses against adversarial examples.
- Currently, different variants of Adversarial Training dominate Robust Bench [3].

References and Resources II

- [1] Anish Athalye, Nicholas Carlini, and David Wagner. **Obfuscated Gradients Give a False Sense of Security: Circumventing Defenses to Adversarial Examples**, 2018. URL <https://arxiv.org/abs/1802.00420>.
- [2] Jacob Buckman, Aurko Roy, Colin Raffel, and Ian Goodfellow. Thermometer Encoding: One Hot Way to Resist Adversarial Examples. In **International Conference on Learning Representations**, 2018. URL <https://openreview.net/forum?id=S18Su--CW>.
- [3] Francesco Croce, Maksym Andriushchenko, Vikash Sehwal, Edoardo Debenedetti, Nicolas Flammarion, Mung Chiang, Prateek Mittal, and Matthias Hein. RobustBench: A Standardized Adversarial Robustness Benchmark, 2021. URL <https://arxiv.org/abs/2010.09670>.
- [4] Guneet S. Dhillon, Kamyar Azizzadenesheli, Zachary C. Lipton, Jeremy Bernstein, Jean Kossaifi, Aran Khanna, and Anima Anandkumar. Stochastic activation pruning for robust adversarial defense, 2018. URL <https://arxiv.org/abs/1803.01442>.
- [5] Ian J. Goodfellow, Jonathon Shlens, and Christian Szegedy. Explaining and Harnessing Adversarial Examples, 2015. URL <https://arxiv.org/abs/1412.6572>.

References and Resources III

- [6] Chuan Guo, Mayank Rana, Moustapha Cisse, and Laurens van der Maaten. Countering Adversarial Images using Input Transformations, 2018. URL <https://arxiv.org/abs/1711.00117>.
- [7] Aleksander Madry, Aleksandar Makelov, Ludwig Schmidt, Dimitris Tsipras, and Adrian Vladu. **Towards Deep Learning Models Resistant to Adversarial Attacks**, 2019. URL <https://arxiv.org/abs/1706.06083>.
- [8] Yang Song, Taesup Kim, Sebastian Nowozin, Stefano Ermon, and Nate Kushman. PixelDefend: Leveraging Generative Models to Understand and Defend against Adversarial Examples, 2018. URL <https://arxiv.org/abs/1710.10766>.
- [9] Cihang Xie, Jianyu Wang, Zhishuai Zhang, Zhou Ren, and Alan Yuille. Mitigating Adversarial Effects Through Randomization, 2018. URL <https://arxiv.org/abs/1711.01991>.
- [10] Hongyang Zhang, Yaodong Yu, Jiantao Jiao, Eric P. Xing, Laurent El Ghaoui, and Michael I. Jordan. **Theoretically Principled Trade-off between Robustness and Accuracy**, 2019. URL <https://arxiv.org/abs/1901.08573>.